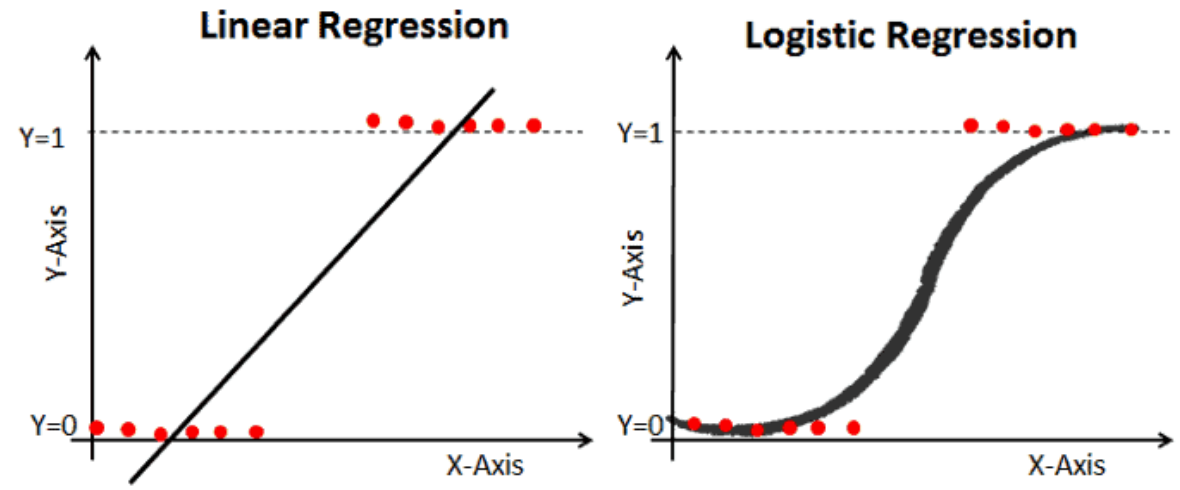
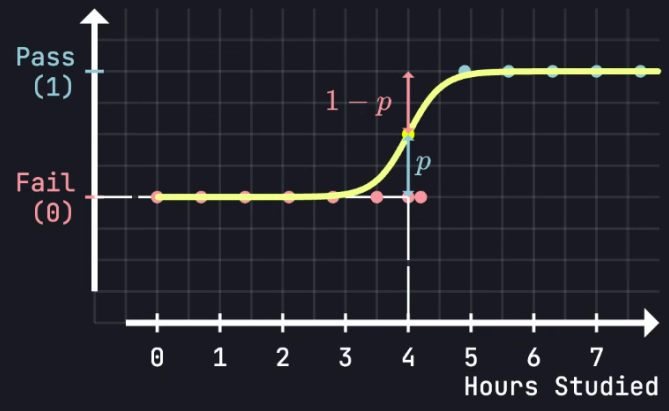
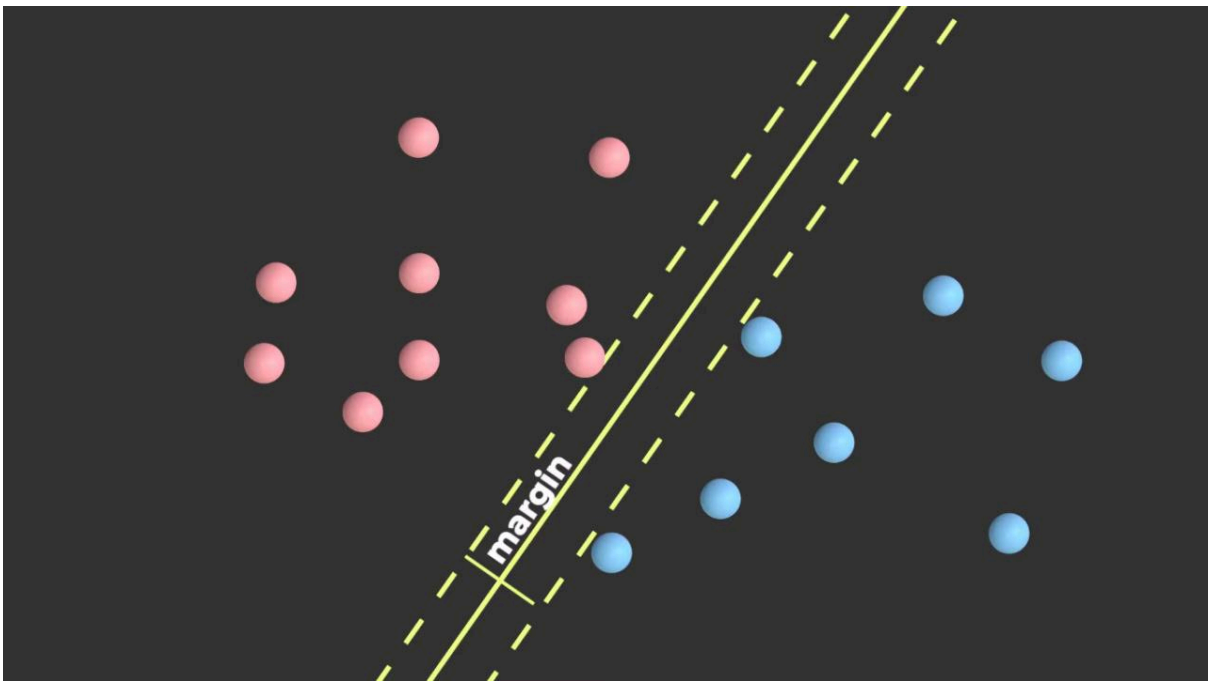
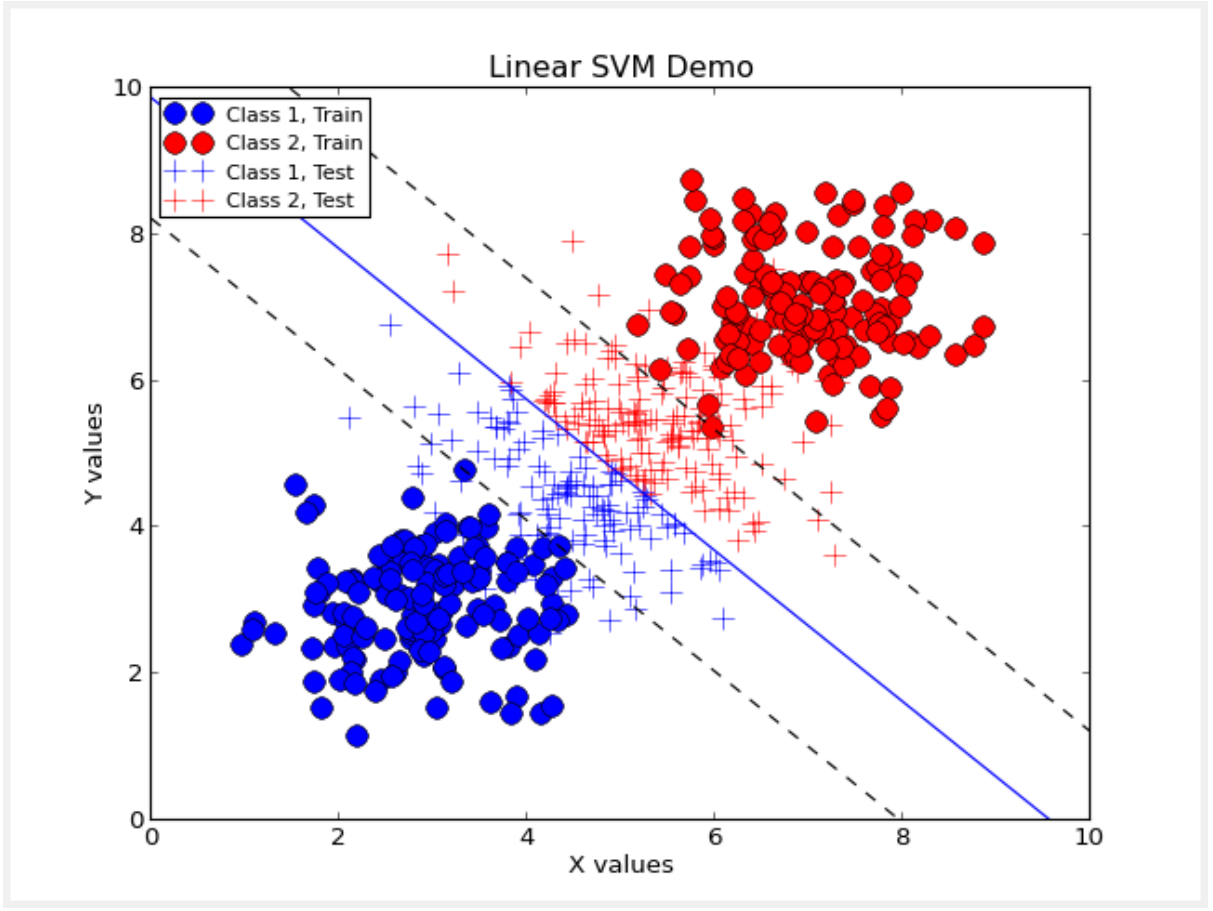
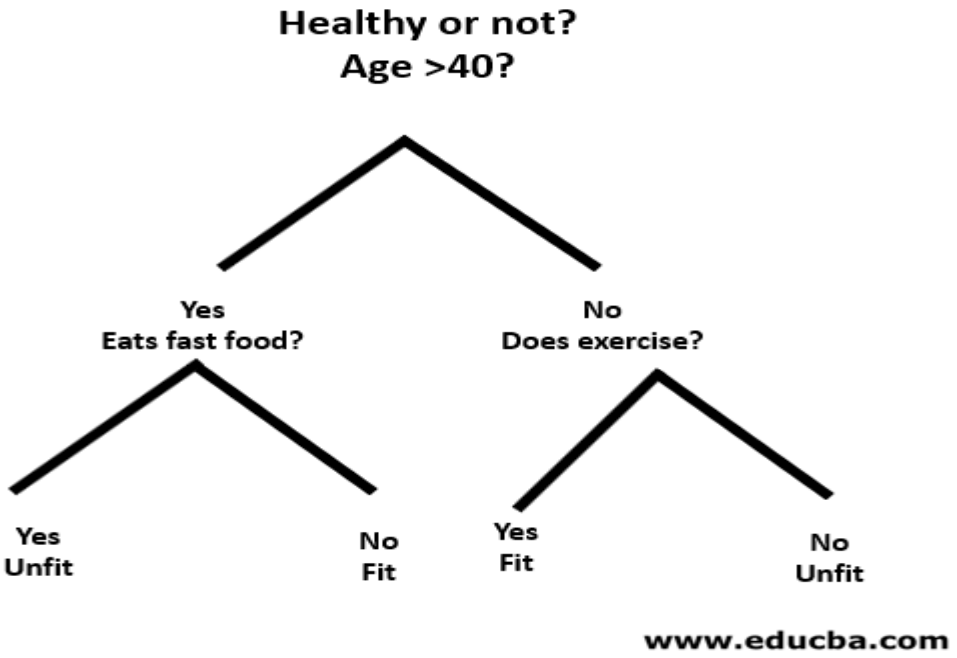


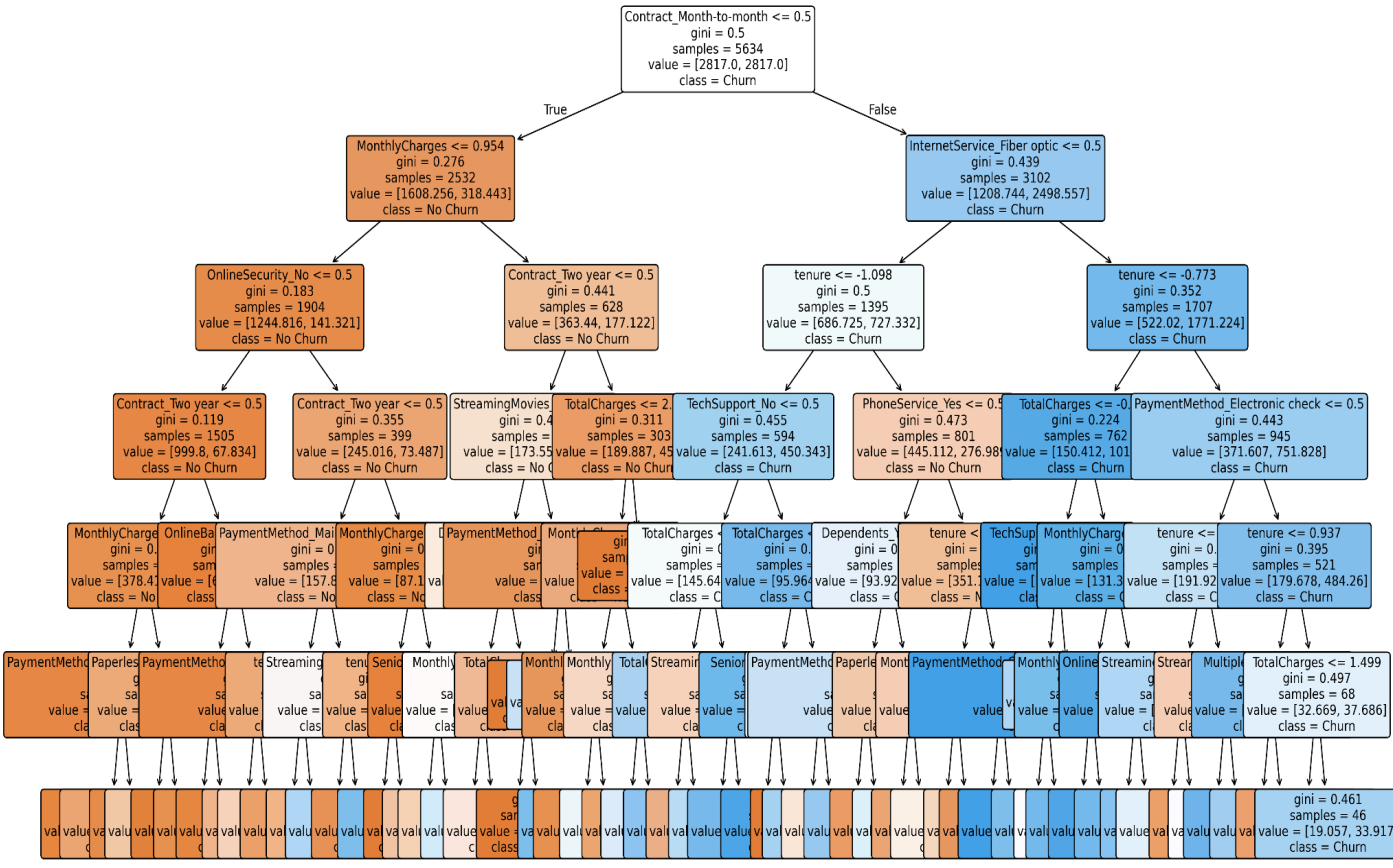
$$-\sum_{\text{Students}} \text{Student Passed?} \times \log(p) + \text{Student Failed?} \times \log(1-p)$$

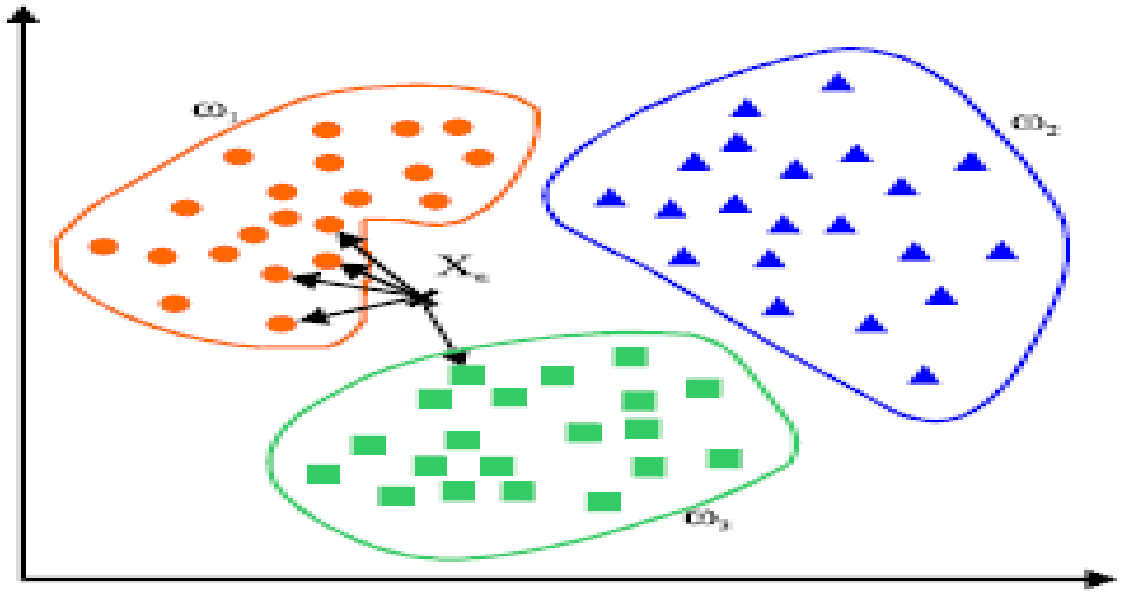


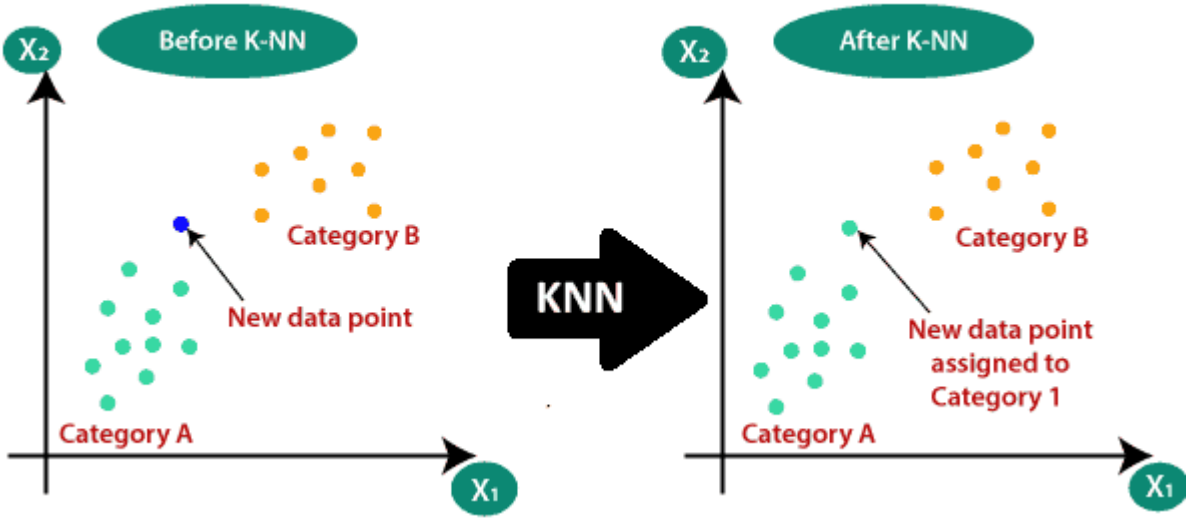

margin


Linear SVM Demo
Class 1, Train
Class 2, Train
Class 1, Test
Class 2, Test
Y values
X values
0
2
4
6
8
10


Healthy or not?
Age >40?
Yes
Eats fast food?
No
Does exercise?
Yes
Unfit
No
Fit
Yes
Fit
No
Unfit
www.educba.com


Contract_Month-to-month <= 0.5
gini = 0.5
samples = 5634
value = [2817.0, 2817.0]
class = Churn
True
False
MonthlyCharges <= 0.954
gini = 0.276
samples = 2532
value = [1608.256, 318.443]
class = No Churn
InternetService_Fiber optic <= 0.5
gini = 0.439
samples = 3102
value = [1208.744, 2498.557]
class = Churn
OnlineSecurity_No <= 0.5
gini = 0.183
samples = 1904
value = [1244.816, 141.321]
class = No Churn
Contract_Two year <= 0.5
gini = 0.441
samples = 628
value = [363.44, 177.122]
class = No Churn
tenure <= -1.098
gini = 0.5
samples = 1395
value = [686.725, 727.332]
class = Churn
tenure <= -0.773
gini = 0.352
samples = 1707
value = [522.02, 1771.224]
class = Churn
Contract_Two year <= 0.5
gini = 0.119
samples = 1505
value = [999.8, 67.834]
class = No Churn
Contract_Two year <= 0.5
gini = 0.355
samples = 399
value = [245.016, 73.487]
class = No Churn
TechSupport_No <= 0.5
gini = 0.455
samples = 594
value = [241.613, 450.343]
class = Churn
TotalCharges <= -0.
gini = 0.224
samples = 762
value = [150.412, 101
class = Churn
PaymentMethod_Electronic check <= 0.5
gini = 0.443
samples = 945
value = [371.607, 751.828]
class = Churn
tenure <= 0.937
gini = 0.395
samples = 521
value = [179.678, 484.26]
class = Churn
TotalCharges <= 1.499
gini = 0.497
samples = 68
value = [32.669, 37.686]
class = Churn
gini = 0.461
samples = 46
value = [19.057, 33.917]
class = Churn




X2
Before K-NN
Category B
New data point
Category A
X1
KNN
X2
After K-NN
Category B
New data point
assigned to
Category 1
Category A
X1